

PREPARED BY **FED** FOR THE MINISTRY OF ELECTRONICS & INFORMATION TECHNOLOGY

# How HStat counts trade

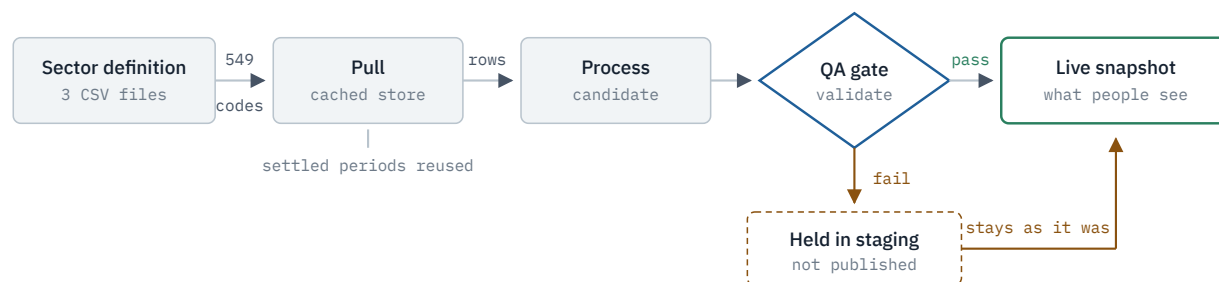
Every figure on the dashboard comes from one source — UN Comtrade’s record of what customs authorities filed — put through one procedure. This sheet sets out that procedure: what the data actually is, how the single global trade figure is built, how re-imports are removed from it, and what keeps it current.

## The approach

QUESTION 1

The dashboard never queries UN Comtrade while someone is looking at it. It reads a stored snapshot that has already been built, checked and approved.

That is a deliberate constraint rather than a technical convenience. A dashboard that queries live cannot refuse to show a bad number, because the number arrives at the same moment the page does. Separating the two gives us a place to stand between the source and the reader — somewhere a figure can be examined and, if it does not hold up, withheld. It also means any figure can be reproduced exactly: the snapshot that produced it is still on disk.



The five stages of a refresh. Nothing reaches the live snapshot without passing the gate, and a failure is not a blank dashboard — the previous validated snapshot simply keeps serving. Only two snapshots are ever kept, **current** and **previous**.

Which HS codes exist is decided by three hand-editable CSV files, not by anything the pipeline discovers at run time. That keeps the scope auditable: the sector definition is a document someone at FED owns and can be asked to justify, rather than an emergent property of a script. 423 rows from the FED Electronics Sector Definition workbook produce 418 current products, and those roll up into the 106 four-digit headings and 20 chapters that contain them — 549 codes pulled in all.

## How the data is interpreted

QUESTION 2

UN Comtrade is not a finished dataset. It is a rolling collection of what individual customs authorities have filed so far, and reading it as though it were complete is the most common way to get trade analysis wrong.

Two properties drive everything that follows. First, a recent period contains whichever economies have reported by now — so summing it and calling the result “the world” understates the total, and the shortfall moves from month to month. A dashboard that did this would appear to report a collapse in trade when it was reporting late paperwork. Second, every shipment is recorded twice, once by the exporter and once by the importer, and the two never agree; that is a property of mirror statistics, not an error.

### Each period is judged against the one before it

A period may be published only if it still contains the economies that mattered last time: the previous largest reporter present, at least 9 of the previous top 10, at least 95% of the previous top-20 value, and a reporter count of at least 80%. Meeting a looser bar (8 of 10, 90% of value, 75% of count) is marked **caution**. Anything else is **invalid**.

### A period that fails is withheld, never estimated

No headline, no rank, no share. The reported observations stay in the underlying data and in the workbook export, so an analyst can still see them — but nothing goes on the page implying a completeness the data does not have.

### India’s own filing must reconcile

India appears twice in the source: in its own bilateral return, and as one row inside the all-reporters frame. The two describe the same trade. If they differ by more than 1%, one of the two extracts is stale and the whole period is demoted rather than published.

### HS 2022 is the base; older codes are history, not products

The workbook mixes classification vintages. 851712 covered all cellular handsets until HS 2022 split it into 851713 smartphones and 851714 others. The retired code gets no page of its own; it is recorded as the successors’ classification history and its data still feeds the four-digit heading.

**Why a split series is never joined up.** The old code’s total covers every successor at once. Dividing it between them requires a share nobody has published, and an invented share would be indistinguishable from measurement. So the six-digit series genuinely begins in 2022, with the old code drawn beside it as a separate dashed line. The continuous long series lives at the four-digit heading, where the split is internal — and because headings are pulled directly from Comtrade, that continuity requires no reconstruction at all.

## The global trade figure

QUESTION 3

The previous version of the dashboard showed reported global imports next to reported global exports. Those are the same trade measured from opposite ends; they never match, and to a reader the difference looks like an error. There is now one number.

$$\text{global trade } (h, t) = \sum_r (M_r - RM_r)$$

for HS code **h** in period **t**, summed over every reporting economy **r**, where **M** is that economy's imports from the world and **RM** is the re-imports it filed separately.

The import side is the headline for two reasons. It is the side that matches how import-substitution and PLI analysis is framed — the question is usually what India buys and from whom. And it is the side with the better coverage: an economy that imports has every incentive to record it at customs, because duty is assessed on it.

The trade-off is that imports are valued **CIF** — cost, insurance and freight — so they sit slightly above the exporter's FOB valuation of the same goods. That is why the export side is still computed and published as a reconciliation gap rather than discarded.

| WORKED EXAMPLE — ONE HS CODE, ONE YEAR |                    |                  |         |
|----------------------------------------|--------------------|------------------|---------|
| ILLUSTRATIVE FIGURES                   |                    |                  |         |
| REPORTING ECONOMY                      | IMPORTS FROM WORLD | RE-IMPORTS FILED | COUNTED |
| China                                  | 120.0              | 4.0              | 116.0   |
| China, Hong Kong SAR                   | 60.0               | 22.0             | 38.0    |
| Singapore                              | 45.0               | 12.0             | 33.0    |
| Germany                                | 20.0               | <i>not filed</i> | 20.0    |
| India                                  | 8.0                | 0.2              | 7.8     |
| World                                  | 253.0              | 38.2             | 214.8   |

Germany in this example did not file re-imports as a separate flow, so its 20.0 is counted as filed. That is what the adjustment-coverage figure exists to disclose: 92% of the world total could be adjusted, the remaining 8% could not, and the true figure is therefore at or slightly below the one shown. Every published figure carries this number, and the product page states it in words.

| WHAT THE DASHBOARD THEN PUBLISHES |       |                                                                    |
|-----------------------------------|-------|--------------------------------------------------------------------|
| Global trade                      | 214.8 | world imports, net of re-imports                                   |
| India share                       | 3.6%  | 7.8 ÷ 214.8                                                        |
| India rank                        | 5th   | on the netted values, not the gross ones                           |
| Adjustment coverage               | 92%   | 233.0 of 253.0 came from reporters that file re-imports separately |
| Mirror gap                        | +6.3% | import side vs. export side, net of re-exports                     |

## Adjusting for re-imports and re-exports

QUESTION 4

Total imports and total exports as filed already contain the re-traded flows. Adding them up without removing those flows counts the same goods more than once.

| UN COMTRADE FLOW ALGEBRA |                     |               |
|--------------------------|---------------------|---------------|
| TOTAL FLOW               | IS MADE OF          | HSTAT REMOVES |
| M imports                | FM + RM + MIP + MOP | RM re-imports |
| X exports                | DX + RX + XIP + XOP | RX re-exports |

A re-import is a good that left an economy and came back — returned stock, goods sent out for repair, an entrepôt movement. It crosses the border twice and is recorded twice, but it is one shipment of one good. Hong Kong and Singapore in the example above are exactly this case: a large share of what they record as imports is destined to leave again.

The removal is done **economy by economy**, not as a single global deduction, because each reporter files its own sub-flow. Three rules govern it:

### Only what is filed is removed

Where a reporter does not break out re-imports, its total is counted as filed and the shortfall is disclosed rather than modelled. No economy has a re-import figure imputed to it.

### An inconsistent filing is refused, not negated

If a reporter files re-imports larger than the total they belong to, the adjustment is declined for that economy and it is flagged. Subtracting anyway would produce a negative contribution to a world total — arithmetically valid, economically meaningless.

### The export side is netted too, as a check

World exports less re-exports is computed alongside, and the ratio between the two sides is published as the mirror gap. A modest positive gap is expected and healthy — CIF includes freight and insurance that FOB excludes. A gap outside roughly  $\pm 30\%$  is recorded as a QA warning against that code and period.

**One limit worth stating plainly.** India's country-by-country breakdown is *not* netted, because re-imports are not filed by partner — no authority records that a particular return shipment came back from a particular country. So supplier shares and concentration measures are computed on the gross basis. This is recorded in the data as basis: "gross" and stated on the page rather than left for the reader to discover.

# India's own tariff lines, and rupees

QUESTION 5

Comtrade stops at six digits. India's eight-digit detail exists only in India's own publication, on a different calendar, and often in a different currency. All three differences are kept visible rather than smoothed away.

Digits seven and eight are a national extension — ITC(HS) for India — and every country's extension means something different, so there is no such thing as a global comparison at that level. Tariff-line detail is therefore a second, India-only source sitting beneath the six-digit page, not a deeper zoom on the same data. It comes from a DGCIS / TradeStat export supplied as a static file.

| TWO SOURCES, TWO CALENDARS, TWO CURRENCIES |                         |                              |
|--------------------------------------------|-------------------------|------------------------------|
|                                            | UN COMTRADE             | DGCIS TARIFF LINES           |
| Detail                                     | HS six digits           | ITC(HS) eight digits         |
| Period                                     | Calendar year, Jan–Dec  | Financial year, Apr–Mar      |
| Coverage                                   | Every reporting economy | India only                   |
| Filed in                                   | US dollars              | Rupees or dollars, by export |

## The two calendars are never merged

Tariff lines are stored in their own block, outside the calendar-year series, and the quality gate fails any snapshot that files a financial year under a calendar year. A bare four-digit year in the input is refused rather than placed: “2024” could mean either financial year, and choosing one silently would shift a year of trade by up to twelve months. On the page the two appear side by side, each labelled, never summed.

## A part year says so, and does not become the default

An export taken mid-year is useful but is not comparable with a full year. The file records how many months it covers; the page marks the period incomplete, prefers the last complete financial year as the default view, and skips the reconciliation check entirely — nine months against twelve tells you nothing.

## Each period converts at its own rate, and only what was filed is shown

Every figure is stored in dollars, because that is how both sources are denominated. A single fixed rate across thirty years would turn an exchange-rate movement into an apparent trade movement, so calendar years use calendar-year rates and financial years use financial-year rates, following the RBI annual average reference rate FED already uses elsewhere; the rate is printed beside the figure it produced. A value published in rupees is shown exactly as filed rather than round-tripped, and the other currency is labelled as derived. Where a period has no sourced rate nothing is converted: the figure stays in dollars and the page says so.

**Where the rupee stops.** Conversion applies to India's own figures and the tariff lines. Global trade, economy rankings and supplier tables stay in dollars — an Indian reference rate is the right way to read an Indian customs filing and the wrong way to read anyone else's.

## Headings, and what they do not cover

QUESTION 6

The four-digit and two-digit figures are Comtrade's own aggregates, pulled at those levels. They cover the whole heading, including six-digit lines this sector definition does not track — so each parent page also states how large a slice it is looking at.

Deriving a heading by adding up the tracked six-digit lines would have been cheaper, and would have been wrong twice over. It would publish a partial figure under the heading's own name: of the 106 headings in scope only 30 are fully covered, and the definition holds 418 of the 805 official six-digit lines inside them. And it would destroy the long series, which is continuous across the 2022 revision precisely because headings are pulled whole — the smartphone split is internal to heading 8517 and needs no reconstruction. A derived heading inherits every split.

| WHAT A SUMMED HEADING WOULD HAVE SHOWN |         |                |         |
|----------------------------------------|---------|----------------|---------|
| HEADING                                | TRACKED | OFFICIAL LINES | COVERED |
| 8501 motors and generators             | 1       | 17             | 5.9%    |
| 3920 plastic sheet and film            | 1       | 22             | 4.5%    |
| 8517 telephony                         | 10      | 23             | 43.5%   |
| All 106 headings                       | 418     | 805            | 51.9%   |

So the official figure stays the published figure, and every parent page carries a coverage panel alongside it: how many of the heading's official lines the definition tracks, what those lines come to, and what share of the heading's trade and of India's imports that is. The denominator is Comtrade's own reference table, kept in the repository so the count does not depend on being online. Where a tracked line had no published figure in a given year the sum omits it and the panel says the share is a lower bound.

**The cost of doing it properly.** Pulling the headings rather than deriving them costs about 124 extra calls on a full rebuild and eight a month thereafter, against an allowance of 500 a day. The accuracy is not being bought at a price worth discussing.

## What keeps it current

QUESTION 7

The dashboard refreshes itself on a schedule, and the schedule is shaped by how fast each part of the data actually moves.

Comtrade revises continuously: an economy that filed late, or corrected a return, changes a figure that was published months ago. But it revises unevenly. The current year and the last few months change constantly; a year three back barely moves at all. Refreshing both on the same cadence would spend most of the effort re-fetching data that has not changed.

| REFRESH CADENCE |                               | RE - FETCHES                | API CALLS |
|-----------------|-------------------------------|-----------------------------|-----------|
| RUN             | WHEN                          |                             |           |
| Light           | Eight months a year           | Current year, last 3 months | 36        |
| Deep            | January, April, July, October | Last 3 years, last 6 months | 90        |
| Full rebuild    | On demand                     | Everything, 1996 onward     | 542       |

Everything already fetched is kept in a local store, so a settled period is retrieved once and served from disk from then on. That is what holds the ongoing cost to roughly 54 calls a month against a 500-a-day allowance — and it is why extending the history back to 1996 was affordable: depth costs one larger first build and nothing thereafter.

### Every stage fails closed

If the pull is incomplete, or the quality checks find a fault, the candidate is held and the live snapshot is left exactly as it was. A stale but validated dashboard is preferable to a fresh but wrong one, and a partial snapshot is the dangerous case — it is missing periods rather than visibly broken.

### The checks are arithmetic, not judgement

Balances must equal exports less imports; netted totals must equal gross less what was removed; shares must fall between 0 and 1; rankings must be ordered and sequential; no headline may exist for a period that failed coverage. A snapshot violating any of these cannot be promoted, by construction rather than by review.

### Scope changes are a file edit, not a code change

Adding an HS code, renaming a product, adjusting the search vocabulary or recording a classification change are all edits to a CSV. The next refresh picks them up. Nobody needs to touch the pipeline to change what the dashboard covers.

Source: UN Comtrade, HS 2022 (H6) at six digits, with official four-digit and two-digit aggregates pulled directly at those levels. India tariff-line detail at ITC(HS) eight digits comes from a DGCIS / TradeStat export when supplied, on financial years. Rupee conversion uses the RBI annual average reference rate, financial years from the Economic Survey Statistical Appendix Table 5.4. Coverage denominators come from Comtrade's own H6 reference table. Figures in the worked example are illustrative and chosen to show the arithmetic; they are not observations.

The methodology statement, the coverage thresholds and the adjustment coverage for every published figure travel with the data itself, in `methodology.json` and each product record, so any number on the dashboard can be traced back to the rule that produced it.